

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

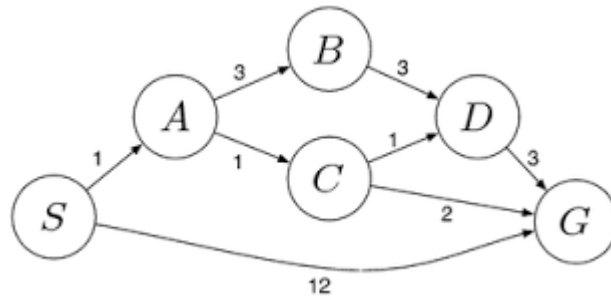
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path

from the starting warehouse **S** to the destination warehouse **G** using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

*I Ganji Madhan Kumar (192472328) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

G.Madhan Kumar

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

1.Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Ans:

Given: 4l gallon,3lgallon.

Intitial state:(0,0)

Goal state:(2,0)

1.Initial state is(0,0).

2.fill the water into 3 liter gallon from the pump(0,3).

3.pour the water into 4 liter gallon(3,0).

4. Now the 3l gallon is empty. Again fill the 3l gallon from the pump(3,3).

5.Again puor the 3l gallon water into the 4l gallon(4,2).

6.Now pour the 4l gallon water on the ground(0,2).

7.Now pour the 3l gallon water into 4l gallon(2,0).

8.Goal state reached(2,0).

Initial State:

4-Gallon = 0 3-Gallon = 0

Fill 3-gallon jug

4-Gallon = 0 3-Gallon = 3

Pour 3-gallon into 4-gallon

4-Gallon = 3 3-Gallon = 0

Fill 3-gallon jug again

4-Gallon = 3 3-Gallon = 3

Pour into 4-gallon until full

4-Gallon = 4 3-Gallon = 2

Empty 4-gallon jug

4-Gallon = 0 3-Gallon = 2

Pour remaining water into 4-gallon jug

4-Gallon = 2 3-Gallon = 0

Goal Achieved!

Exactly 2 gallons are in the 4-gallon jug.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

- i. What are the percept's for this agent?**
- ii. Characterize the operating environment.**
- iii. What are the actions the agent can take?**
- iv. How can one evaluate the performance of the agent?**
- v. What sort of agent architecture do you think is most suitable for this agent? Why?**

i. What are the percepts for this agent?

Percepts are the inputs received by the Mars Rover through its sensors.

- Camera images of the Martian surface
- Rock and soil composition data
- Temperature and weather conditions
- Terrain and obstacle information
- Distance to nearby objects
- Position and orientation (GPS-like localization)
- Battery/power level
- Communication signals from Earth

ii. Characterize the operating environment

The Mars Rover operates in the following environment:

- Partially Observable: Not all information about Mars is available at once.
- Deterministic: Most actions have predictable outcomes, though some uncertainty exists.
- Sequential: Current actions affect future actions.
- Dynamic: Weather (dust storms) and terrain conditions may change.
- Continuous: Time, movement, and sensor readings are continuous.
- Single-Agent: The rover performs tasks independently.

iii. What are the actions the agent can take?

- Move forward, backward, left, or right
- Navigate around obstacles
- Capture images and videos
- Collect rock and soil samples
- Analyze samples using onboard instruments
- Drill rocks

iv. How can one evaluate the performance of the agent?

The Mars Rover's performance can be evaluated based on:

- Accuracy of sample collection and analysis
- Number of scientific discoveries
- Successful navigation without collisions
- Efficient energy (battery) usage
- Reliability of communication with Earth

v. What sort of agent architecture is most suitable for this agent? Why?

Model-Based Goal-Based Agent Architecture

- Maintains an internal model of the environment.
- Plans actions to achieve mission goals.
- Uses sensor data to update its understanding of the surroundings.
- Can adapt to unexpected obstacles and changing terrain.
- Makes intelligent decisions with limited communication from Earth.

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

Ans:

Algorithm Steps

1. Start with an empty board.
2. Place a queen in the first column.
3. Move to the next column and place another queen in a safe position.
4. If no safe position exists, **backtrack** to the previous column and move that queen.
5. Repeat until all 8 queens are safely placed.

code:

```
# Function to print the chessboard
def print_board(board):
    for row in board:
        print(" ".join("Q" if x else "." for x in row))

# Check if placing queen is safe
def is_safe(board, row, col):
    # Check left side of row
    for i in range(col):
        if board[row][i]:
            return False

    # Check upper-left diagonal
    i, j = row, col
    while i >= 0 and j >= 0:
        if board[i][j]:
            return False
        i -= 1
        j -= 1

    # Check lower-left diagonal
    i, j = row, col
    while i < 8 and j >= 0:
        if board[i][j]:
            return False
        i += 1
        j -= 1

    return True

# Solve using backtracking
def solve(board, col):
    if col == 8:
        return True

    for row in range(8):
        if is_safe(board, row, col):
            board[row][col] = 1
```

```

        if solve(board, col + 1):
            return True

        board[row][col] = 0

    return False

# Main program
board = [[0 for _ in range(8)] for _ in range(8)]

if solve(board, 0):
    print("Solution Found:\n")
    print_board(board)
else:
    print("No Solution Exists")

```

Output:

```

Q . . . . . . .
. . . . . Q .
. . . . Q . .
. . . . . . Q
. Q . . . . .
. . . Q . . .
. . . . . Q .
. . Q . . . .

```

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Ans:

Type of Problem-Solving Agent

The Goal-Based Agent is the most suitable agent.

Reason:

- The customer's goal is to reach the destination.
- The agent evaluates different cab options (Mini, Micro, Sedan, Prime, Shared, etc.).
- It selects the cab that best satisfies the customer's preferences such as cost, comfort, and travel time.

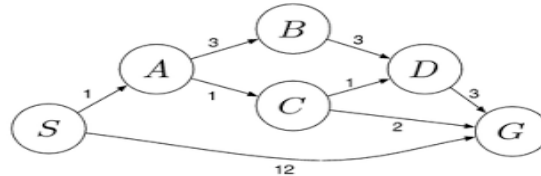
Problem Components

- Initial State: Customer is at the pickup location.
- Goal State: Customer reaches the destination.
- Actions:
 - Enter pickup and destination.
 - Display available cab types.
 - Select a cab.
 - Confirm booking.

- Travel to destination.
- Path Cost: Fare, travel time, and waiting time.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse **G** using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Uniform Cost Search (UCS) Algorithm

Step 1: Start at node **S** with cost 0.

Step 2: Insert **S** into the priority queue.

Step 3: Remove the node with the lowest path cost.

Step 4: If it is the goal node **G**, stop.

Step 5: Otherwise, expand the node and add its neighbors with cumulative costs.

Step 6: Repeat until **G** is reached.

Pseudocode

UniformCostSearch(Graph, Start, Goal)

Create a priority queue **PQ**

Insert (Start, 0) into **PQ**

While **PQ** is not empty

Remove node with minimum cost

If node = Goal

Return path and total cost

For each neighboring node

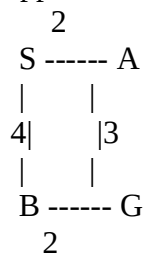
NewCost = CurrentCost + EdgeCost

Insert neighbor with NewCost into **PQ**

End While

Example (Illustration)

Suppose the graph is:



Possible paths:

- $S \rightarrow A \rightarrow G = 2 + 3 = 5$
- $S \rightarrow B \rightarrow G = 4 + 2 = 6$

Uniform Cost Search selects:

Least-cost path: $S \rightarrow A \rightarrow G$

Total Cost: 5